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Maths



NORTH SYDNEY BOYS HIGH SCHOOL

2013 YEAR 11 YEARLY EXAMINATION

Mathematics

General Instructions

- Working time – 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions
- Marks may be deducted for incomplete or illegible working

Class Teacher:

(Please tick or highlight)

☐ Mr Lin

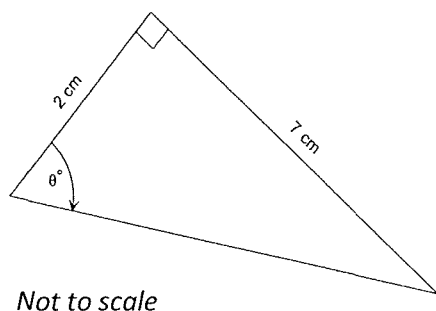
Student Name:

(To be used by the exam markers only.)

Question No	1-7	8	9	10	11	12	13	14	Total
Mark	$\overline{7}$	$\overline{13}$	$\overline{14}$	$\overline{13}$	$\overline{15}$	$\overline{8}$	$\overline{13}$	$\overline{17}$	$\overline{100}$

MULTIPLE CHOICE

1. What is the solution to $x^2 - 6x - 7 < 0$
- (a) $-7 < x < 1$ (b) $-1 < x < 7$
- (c) $x < -7$ or $x > 1$ (d) $x < -1$ or $x > 7$
2. The function $f(x) = \frac{x}{x+1}$ is a:
- (a) Exponential (b) Hyperbola
- (c) Parabola (d) None of the above
3. The value of θ (to the nearest degree) is:



- (a) 16° (b) 17°
- (c) 73° (d) 74°

4. Find the perpendicular distance between $x + y - 1 = 0$ and $(2, -2)$

- (a) $\frac{\sqrt{2}}{2}$ (b) $\frac{2}{\sqrt{2}}$
- (c) $\frac{1}{\sqrt{8}}$ (d) $\frac{\sqrt{2}}{4}$

5. What is the external angle sum of a regular hexagon:

- (a) 60° (b) 120°
- (c) 360° (d) 720°

6. If $w = z^x$, which of the following statements is true?

- (a) $w = \log_z x$ (b) $x = \log_z w$
- (c) $z = \log_w x$ (d) $z = \log_x w$

7. Evaluate: $\lim_{x \rightarrow 2} \frac{4 - 2x}{4 - x^2}$

- (a) 0 (b) $\frac{1}{2}$
- (c) 1 (d) 2

Question 8 – Basic Arithmetic Algebra & Equations (13 Marks)

- (a) Evaluate correct to 3 significant figures: $\frac{3.14 - \sqrt{47 - 4 \times 4 \times 2}}{2^2 - 5^2}$ 2
- (b) Express $1.\overline{234}$ as a fraction 2
- (c) Rationalise the denominator of: $\frac{3}{1 - \sqrt{2}}$ 2
- (d) Solve each of the following equations for x
- (i) $3(x - 2) + x = 2 + 2x$ 2
- (ii) $\frac{3x}{x+1} = \frac{2}{5}$ 2
- (e) Make v the subject of the equation: $wv - 2x = v + 5$ 3

Question 9 – Functions and Graphs (14 Marks)

- (a) Is the function $f(x) = \frac{x}{2+x^2}$ odd, even or neither? Justify your answer with working. 3
- (b) Sketch each of the following on separate number planes showing all key features
- (i) $y = x^2 - 3x - 10$ 3
- (ii) $y = -\sqrt{9 - x^2}$ 2
- (c) On the same number plane clearly sketch:
- (i) $f(x) = 2^x$ 2
- (ii) $f(x + 1)$ 2
- (iii) $f(x) - 1$ 2

Clearly showing at least two points on each graph

Question 10 - Trigonometry (13 Marks)

- (a) In triangle ABC, AC = 8cm, BC = 10cm and $\hat{A}BC = 50^\circ$. Find the value(s) of $\hat{B}AC$ 3
- (b) Two boats leave a port at the same time. Boat A is travelling at 30 km/h on a bearing of $035^\circ T$ while boat B is travelling at 35 km/h on a bearing of $140^\circ T$.
- (i) Draw a clear diagram of the situation in your answer book. 1
- (ii) How far apart are the boats after 4 hours? 3

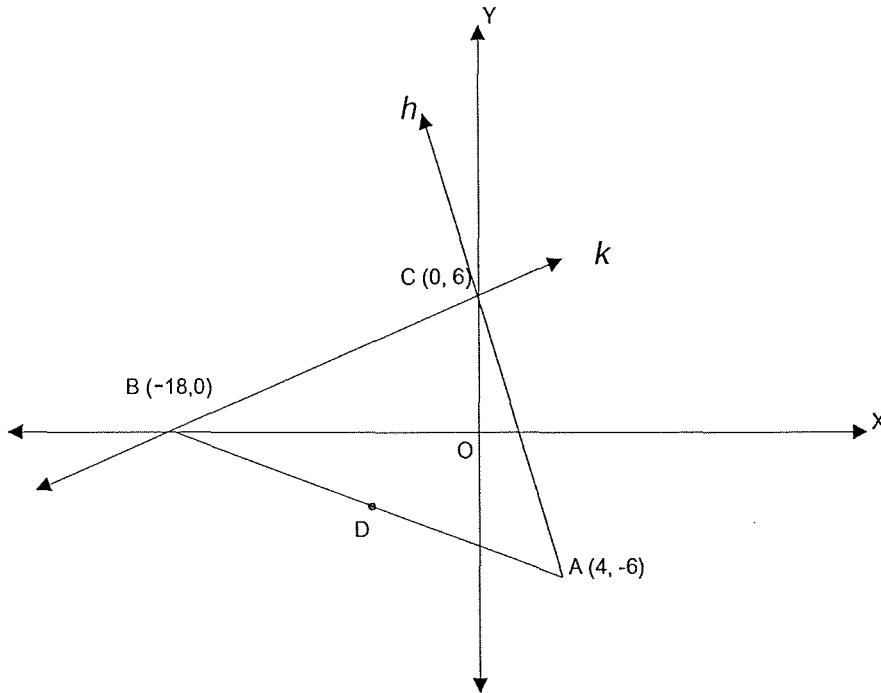
Question 10 continues on the next page

- (c) Given that $\tan x = \frac{1}{4}$ and $\cos x > 0$. Find the exact value of $\operatorname{cosec} x$: 3
- (d) Find all possible solutions for $\sin \theta - \sqrt{3} \cos \theta = 0$ for $0^\circ \leq \theta \leq 90^\circ$ 2
- (e) Show that $\frac{\sin^3 x}{\cos x} + \sin x \cos x = \tan x$ 3

Question 11 - Linear Functions (15 Marks)

- (a) In the diagram the lines h and k are drawn. The coordinates of A, B and C are $(4, -6)$, $(-18, 0)$ and $(0, 6)$ respectively. D is the midpoint of AB .

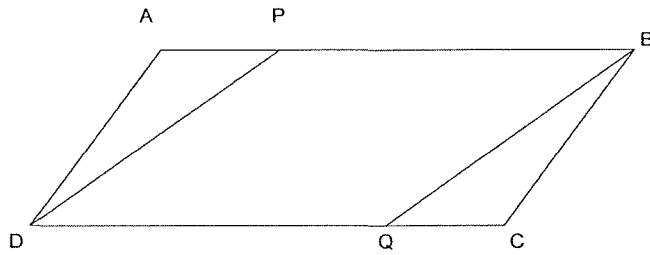
Not to scale



- (i) Show that D has coordinates $(-7, -3)$ 1
- (ii) Calculate the length of DC . Leave your answer in exact form 1
- (iii) Show that the equation of the line h is $3x + y - 6 = 0$ 2
- (iv) Show that line h is perpendicular to line k . 2
- (v) AB is the diameter of a circle which passes through the points A, B, and C. Find the equation of this circle. 2
- (vi) Hence, or otherwise, find the area of the circle through points A, B and C. Leave your answer in terms of π . 1
- (b) (i) Find the point(s) of intersection of: $y = x^2$ and $y = 5x - 4$. 3
- (ii) Hence, indicate on the number plane the region defined by: $y \geq x^2$ and $y < 5x - 4$. 3

Question 12 – Plane (Euclidean) Geometry (8 Marks)

- (a) ABCD is a parallelogram and $AP = QC$.



Prove:

(i) $\triangle APD \equiv \triangle BQC$

(ii) $PD = QB$

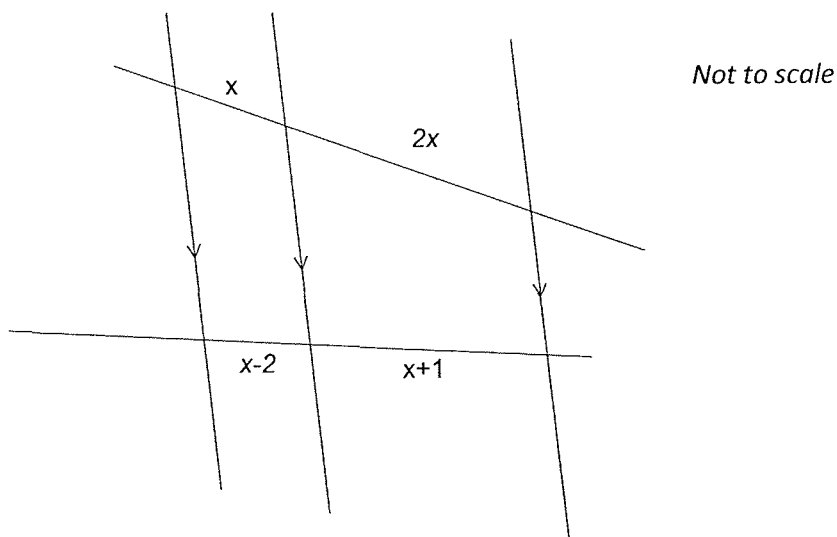
(iii) $PD \parallel QB$

3

1

2

- (b) Find the value of x (give reasons)



2

Question 13 – Indices & Logarithms (13 Marks)

- (a)

Fully simplify $\frac{(x^2 y^{\frac{1}{2}})^2}{y \sqrt{x^3}}$

2

- (b) Find the value of the x when $3^{x-1} = \frac{1}{27}$

2

Question 13 continues on the next page

(c) If $\log_a x = 1.8$ and $\log_a y = 2.4$, evaluate each of the following:

(i) $\log_a xy$ 1

(ii) $\log_a \sqrt[3]{y}$ 1

(iii) $\log_a \frac{a}{x}$ 2

(d) Solve for x : $\log_2(x - 1) - \log_2(x - 2) = 1$ 3

(e) Is the following working correct? Explain your answer.

$$\left(\frac{1}{2}\right)^x < 7$$

$$\log\left(\frac{1}{2}\right)^x < \log 7$$
2

$$x \log\left(\frac{1}{2}\right) < \log 7$$

$$x < \frac{\log 7}{\log \frac{1}{2}}$$

$$\therefore x < -2.8 \text{ (1 d.p.)}$$

Question 14 – Differential Calculus (17 Marks)

(a) Is this function, $f(x)$, continuous at $x = 1$. Justify your answer with working.

$$f(x) = \begin{cases} x^2 + x + 5 & \text{for } x \leq 1 \\ 2x + 3 & \text{for } x > 1 \end{cases}$$
3

(b) Differentiate each of the following with respect to x .

(i) $3x^5 + x^3 - 5x^2 + 20$ 1

(ii) $x^2\sqrt{1+2x}$ 3

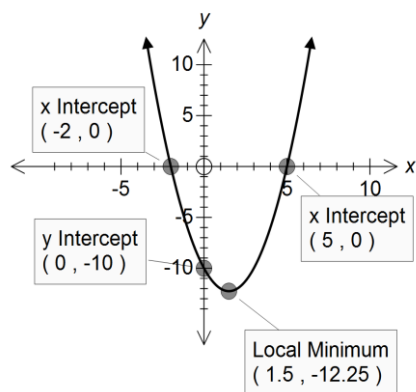
(iii) $\frac{x+2}{2x-1}$ 3

(c) Differentiate $\frac{2}{x}$ using first principles. 3

(d) Calculate the area that the tangent to the curve $y = x^3 - 1$ that passes through the point $(-2, -9)$ makes with the coordinate axes. 4

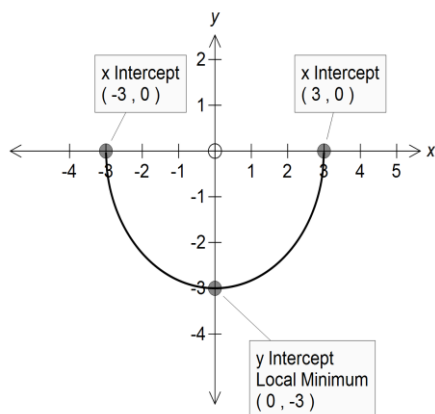
	Multiple Choice	
	1. B 2. B 3. D 4. A 5. C 6. B 7. B	✓ 1 mark each
	Question 8	
(a)	0.0349	✓ Answer ✓ Rounding
(b)	$\text{let } x = 1.234343434 \dots \text{ ①}$ $\therefore 100x = 123.434343 \dots \text{ ②}$ $(2) - (1)$ $99x = 122.2$ $x = \frac{122.2}{99}$ $= \frac{611}{495} \text{ or } 1\frac{116}{49}$	✓ Multiplying by 100 ✓ Correct answer
(c)	$\frac{3}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}} = \frac{3+3\sqrt{2}}{1-2}$ $= -3-3\sqrt{2}$	✓ Multiplying by conjugate ✓ Correct answer
(d) (i)	$3x - 6 + x = 2 + 2x$ $4x - 6 = 2 + 2x$ $2x = 8$ $x = 4$	✓ Expanding brackets and gathering like terms ✓ Correct answer
(d) (ii)	$\frac{3x}{x+1} = \frac{2}{5}$ $15x = 2x + 2$ $13x = 2$ $x = \frac{2}{13}$	✓ Cross multiplying ✓ Correct answer
(e)	$wv - 2x = v + 5$ $wv - v = 2x + 5$ $v(w - 1) = 2x + 5$ $v = \frac{2x + 5}{w - 1}$	✓ All v's to one side ✓ Factorising ✓ Correct answer
	Question 9	
(a)	$f(-x) = \frac{(-x)}{2 + (-x)^2}$ $= \frac{-x}{2 + x^2}$ $= -\frac{x}{2 + x^2}$ $= -f(x)$ $\therefore f(x) \text{ is odd}$	✓ Substituting in $-x$ ✓ Working ✓ Final answer

(b) (i)



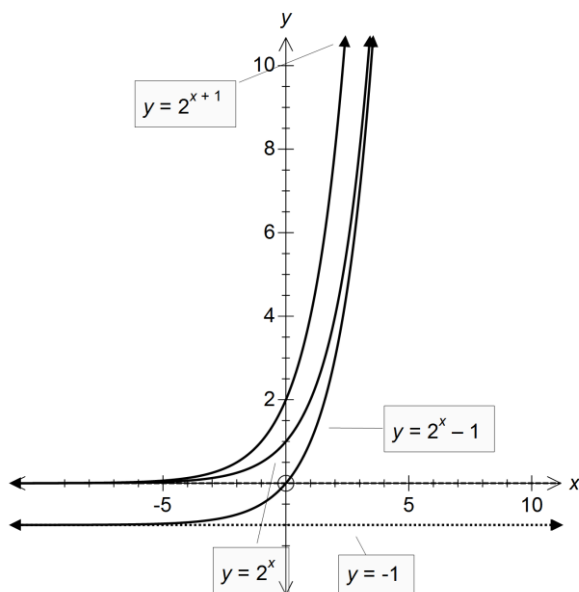
- ✓ X-intercepts
- ✓ Y-intercept
- ✓ Vertex

(b) (ii)



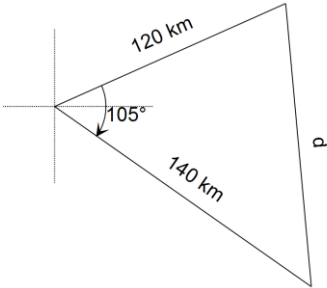
- ✓ Correct shape
- ✓ Correct intercepts

(c)

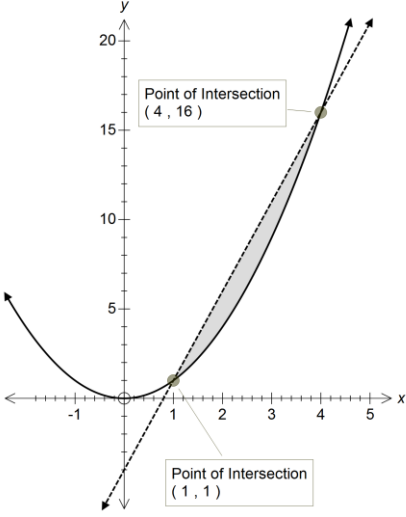


- ✓ Correct shape/(including translation)
- ✓ At least 2 points clearly shown

(times 3 – 2 marks each graph)

	Question 10	
(a)	let $\angle BAC = \theta$ $\frac{\sin \theta}{10} = \frac{\sin 50}{8}$ $\sin \theta = \frac{10 \sin 50}{8}$ $\theta = \sin^{-1}\left(\frac{10 \sin 50}{8}\right)$ $\therefore \theta = 73^\circ$ or $\theta = 180^\circ - 73^\circ$ $= 107^\circ$	✓ Applying sine rule ✓ Acute answer ✓ Obtuse answer
(b) (i)		✓ diagram
(b) (ii)	$d^2 = 120^2 + 140^2 - 2 \times 120 \times 140 \times \cos 105^\circ$ $\therefore d = 207 \text{ km (nearest km)}$	✓ cosine rule ✓ correct substitution ✓ correct answer
(c)	hypotenuse $= \sqrt{4^2 + 1^2}$ $= \sqrt{17}$ $\operatorname{cosec} x = \frac{\text{hypotenuse}}{\text{opposite}}$ $= + \frac{\sqrt{17}}{1} \text{ (first quadrant)}$ $= \sqrt{17}$	✓ Pythagoras theorem ✓ Correct ratio for cosec ✓ Correct sign/final answer
(d)	$\sin \theta = \sqrt{3} \cos \theta$ $\tan \theta = \sqrt{3}$ $\theta = \tan^{-1} \sqrt{3}$ $\theta = 60^\circ$	✓ Equation for tan ✓ Correct answer

(e)	$LHS = \frac{\sin^3 x}{\cos x} + \sin x \cos x$ $= \frac{\sin^3 x}{\cos x} + \frac{\sin x \cos^2 x}{\cos x}$ $= \frac{\sin x (\cancel{\sin^2 x} + \cos^2 x)}{\cos x}$ $= \frac{\sin x}{\cos x}$ $= RHS$	✓ Common denominator ✓ $\sin x + \cos x = 1$ ✓ final answer
	Question 11	
(a) (i)	$D = \left(\frac{-18 + 4}{2}, \frac{0 - 6}{2} \right)$ $= (-7, -3)$	✓ Correct working
(a)(ii)	$DC = \sqrt{(-7 - 0)^2 + (-3 - 6)^2}$ $= \sqrt{49 + 81}$ $= \sqrt{130}$	✓ Correct working
(a)(iii)	$\frac{y - 6}{x - 0} = \frac{-6 - 6}{4 - 0}$ $\frac{y - 6}{x} = -\frac{12}{4}$ $\frac{y - 6}{x} = -3$ $y - 6 = -3x$ $\therefore 3x + y - 6 = 0$	✓ Correct substitution into formula ✓ Adequate working
(a)(iv)	$m_h \times m_k = \frac{-6 - 6}{4 - 0} \times \frac{6 - 0}{0 + 18}$ $= -\frac{12}{4} \times \frac{6}{18}$ $= -\frac{72}{72}$ $= -1$ $\therefore$ the lines are perpendicular	✓ Calculating gradient ✓ Showing gradients multiple to -1
(a) (v)	$(x + 7) + (y + 3) = 130$	✓ Centre ✓ radius
(a) (vi)	$= 130\pi$	✓ Correct answer

(b)(i)	$x^2 = 5x - 4$ $x^2 - 5x + 4 = 0$ $(x - 1)(x - 4) = 0$ $x = 1 \text{ or } x = 4$ $y = 1^2 \text{ or } y = 4^2$ $(1, 1) \text{ and } (4, 16)$	✓ Solving simultaneously ✓ x coordinates ✓ y coordinates
(b) (ii)		✓ graphs correct ✓ shaded region ✓ open circles at points of intersection
Question 12		
(a) (i)	<i>In Δs APD & BQC</i> $AD = CB$ (<i>opposite sides of a parallelogram are equal</i>) $\angle DAP = \angle BCP$ (<i>opposite angles of a parallelogram are equal</i>) $AP = QC$ (<i>given</i>) $\therefore \Delta APD \equiv \Delta BQC$ (SAS)	✓ equal sides and angle identified ✓ all reasons given ✓ final line including SAS
(a) (ii)	$PD = QB$ (matching sides in congruent triangles are equal)	✓ Correct reason
(a) (iii)	$AP = QC$ (<i>given</i>) $AB = CD$ (<i>opposite sides of parallelogram ABCD</i>) $\therefore AB - AP = CD - QC$ $\therefore PB = QC$ $\therefore PBQD$ is a parallelogram (<i>two pairs of opposite sides equal</i>) $\therefore PD \parallel QB$ (<i>opposite sides of a parallelogram are parallel</i>)	✓ $PB = QC$ ✓ Correct reason

(b)	$\frac{x}{2x} = \frac{x-2}{x+1} \text{ (ratio of intercepts on parallel lines)}$ $\frac{1}{2} = \frac{x-2}{x+1} \quad (x \neq 0)$ $2x - 4 = x + 1$ $x = 5$	✓ Ratio of intercepts (must state reason) ✓ Correct answer
Question 13		
(a)	$\frac{\left(x^2 y^{\frac{1}{2}}\right)^2}{y \sqrt{x^3}} = \frac{x^4 y}{x^{\frac{3}{2}} y}$ $= x^{4 - \frac{3}{2}}$ $= x^{\frac{5}{2}} \text{ or } \sqrt{x^5}$	✓ Simplifying numerator and denominator individually ✓ Final answer
(b)	$3^{x-1} = 3^{-3}$ $x - 1 = -3$ $x = -2$	✓ Expressing RHS with a negative index ✓ answer
(c) (i)	$\log_a xy = \log_a x + \log_a y$ $= 1.8 + 2.4$ $= 4.2$	✓ Correct answer
(c) (ii)	$\log_a \sqrt[3]{y} = \log_a (y)^{\frac{1}{3}}$ $= \frac{1}{3} \log_a y$ $= \frac{1}{3} \times 2.4$ $= 0.8$	✓ Correct answer
(c) (iii)	$\log_a \frac{a}{x} = \log_a a - \log_a x$ $= 1 - 1.8$ $= -0.8$	✓ Simplifying log ✓ Correct answer
(d)	$\log_2(x-1) - \log_2(x-2) = 1$ $\log_2\left(\frac{x-1}{x-2}\right) = \log_2 2$ $\therefore \frac{x-1}{x-2} = 2$ $x-1 = 2x-4$ $x = 3$	✓ Simplifying LHS ✓ Writing RHS in terms of a log ✓ Correct answer
(e)	It is incorrect since the inequality on the 4 th lines should be turned the other way as we are dividing by $\log \frac{1}{2}$, which is negative.	✓ Stating it is incorrect ✓ Identifying the reason

Question 14		
(a)	$\lim_{x \rightarrow 1^-} f(x) = 1^2 + 1 + 5$ $= 7$ $\lim_{x \rightarrow 1^+} f(x) = 2(1) + 3$ $= 5$ $\therefore \lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$ $\therefore \text{function is not continuous when } x=1$	✓ Evaluating limits ✓ Correct notation ✓ Stating it is discontinuous
(b) (i)	$15 + 3x - 10$	✓ Correct answer
(b) (ii)	$\frac{d}{dx}(x^2\sqrt{1+2x}) = \left(\frac{d}{dx}(x^2)\right)(\sqrt{1+2x}) + \left(\frac{d}{dx}(\sqrt{1+2x})\right)(x^2)$ $= (2x)(1+2x)^{\frac{1}{2}} + \left((2)\left(\frac{1}{2}\right)(1+2x)^{-\frac{1}{2}}\right)(x^2)$ $= 2x\sqrt{1+2x} + \frac{x^2}{\sqrt{1+2x}}$	✓ Chain rule ✓ Product rule ✓ Answer
(b) (iii)	$\frac{d}{dx}\left(\frac{x+2}{2x-1}\right) = \frac{\left(\frac{d}{dx}(x+2)\right)(2x-1) - \left(\frac{d}{dx}(2x-1)\right)(x+2)}{(2x-1)^2}$ $= \frac{(1)(2x-1) - (2)(x+2)}{(2x-1)^2}$ $= \frac{2x-1-2x-4}{(2x-1)^2}$ $= \frac{-5}{(2x-1)^2}$	✓ Knowing quotient rule ✓ Correct substitutions into formula ✓ Correct answer
(c)	$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2}{x+h} - \frac{2}{x}}{h}$ $= \lim_{h \rightarrow 0} \frac{\left(\frac{2x - 2(x+h)}{x(x+h)}\right)}{h}$ $= \lim_{h \rightarrow 0} \frac{\left(\frac{2x - 2x - 2h}{x(x+h)}\right)}{h}$ $= \lim_{h \rightarrow 0} \frac{\left(\frac{-2h}{x(x+h)}\right)}{h}$ $= \lim_{h \rightarrow 0} \frac{-2}{x(x+h)}$ $= -\frac{2}{x^2}$	✓ Correct use of formula ✓ Cancelling the h's ✓ Answer

(d)	$\frac{dy}{dx} = 3x^2$ $\therefore m_{tgt} = 3(-2)^2$ $= 12$ $y - y_1 = m(x - x_1)$ $y + 9 = 12(x + 2)$ $y + 9 = 12x + 24$ $y = 12x + 15$ $\therefore y\text{-int} = 15$ when $y = 0$ $12x = -15$ $x = -\frac{15}{12}$ $\therefore x\text{-int} = -1.25$ $\therefore \text{Area} = \frac{1}{2} \times 15 \times -1.25 $ $= \frac{75}{8} \text{ units}^2$	✓ gradient of tangent ✓ Equation of tangent ✓ x & y intercepts ✓ area
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